

Project 2 Teacher Reflection Lab 2.0: Design Report

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1. Introduction: Learning Goal and Product Description

Teacher Reflection Lab 2.0 is a web-based platform designed to help K-12 teachers better understand their own teaching style and practice thoughtful responses to complex classroom situations. The platform addresses a recognized gap in teacher professional development: most training programs focus on curriculum delivery and classroom management, while the personal and interpersonal dimensions of teaching, such as how a teacher naturally tends to respond to student conflict, unexpected contributions, or emotional escalation, receive far less systematic attention. The Teacher Reflection Lab 2.0 aims to make those tendencies visible and practicable.

The current prototype consists of two connected components. The first is a Big Five personality self-assessment adapted specifically for teaching contexts. Rather than using generic trait statements, the 20-question quiz presents classroom-specific dilemmas as prompts, scoring responses across the five Big Five dimensions (Openness, Conscientiousness, Extraversion, Agreeableness, and Emotional Stability; John & Srivastava, 1999). The scores are then combined through weighted formulas to generate one of six teacher identity types: Community Energizer, Structured Guide, Creative Explorer, Steady Supporter, Reflective Designer, and Quiet Specialist. Each type presents an identity overview, a set of natural strengths, and targeted growth edges written in accessible, strengths-based language. Teachers may retake the assessment at any time to track how their self-perceptions shift.

The second component is a classroom scenario practice system. Teachers engage with branching narrative simulations built from documented classroom transcripts, making facilitation decisions at three key moments in a real teaching situation. Each decision point offers one weak move, one partial move, and one strong concept-building move; choices map to one of four performance levels (Failure, Partial Success, Strong Partial Success, or Success) and lead to a coaching debrief tailored to the specific path taken. The current prototype demonstrates this system using Sean's Number case (Mathematics Teaching and Learning to Teach, 2010), a third-grade mathematics discussion in which a student argues that six can be both odd and even. The platform is designed to expand into a broader library of cases across grade levels and subject areas, with scenario recommendations matched to each teacher's identity type and self-identified areas of growth.

2. Intended Users and Deployment Context

Teacher Reflection Lab 2.0 is designed for K-12 teachers at any career stage who are seeking a structured space for professional self-reflection. This includes pre-service teachers completing practicum placements, who are beginning to encounter the gap between theoretical knowledge and live classroom decision-making; early-career teachers developing their facilitation skills without consistent access to mentoring; and experienced teachers who want a low-stakes environment to examine habitual responses to challenging classroom moments.

The platform is intended to function as a personal professional development tool rather than an institutional assessment instrument. Teacher professional development is frequently organized around external evaluation: observations, instructional coaching, and administrator feedback, which can constrain honest self-reflection. Research on teacher well-being underscores why this

matters: the 2024 State of the American Teacher Survey found that 44 percent of teachers reported frequent job-related stress, with burnout rates considerably higher than those of comparable working adults (Doan et al., 2024). A self-directed tool that teachers use on their own terms, without results being shared with administrators or evaluators, creates conditions for more genuine engagement with professional growth.

In terms of deployment, the platform is most naturally used as a self-initiated activity accessible from any browser without institutional login. Teachers might engage with it during a planning period, before or after a challenging week, or as part of a regular reflective habit. It could also be introduced by teacher educators in preparation programs as an optional companion resource. The personality results and scenario debrief content belong to the teacher; what they choose to discuss with colleagues or instructors is entirely their own decision.

3. How the Design Supports Learning

The design of Teacher Reflection Lab 2.0 integrates three theoretical foundations: personality self-awareness as a reflective scaffold, scenario-based situated practice, and formative coaching feedback. Two rounds of playtesting shaped the development of each component. Accessibility was also a deliberate design consideration throughout.

Personality self-awareness as a reflective scaffold. Research on teacher personality provides the theoretical grounding for the assessment component. Kim et al. (2019), in a meta-analysis of 25 studies ($N = 6,294$), found that teacher personality traits, including openness, conscientiousness, extraversion, and emotional stability, were positively associated with teacher effectiveness, and that emotional stability, extraversion, and conscientiousness were negatively associated with burnout. Perera et al. (2018) identified four latent personality profiles among 574 Australian teachers and showed that these profiles predicted meaningful differences in self-efficacy, work engagement, and job satisfaction. Research on teacher personality domains further indicates that while Big Five traits show some stability, the behavioral and emotional skills associated with those traits are developable through deliberate practice (Kim et al., 2019). This underpins the platform's core design stance: the identity type is a starting point for reflection, not a fixed description of the teacher.

The personality component underwent a significant revision driven by the first round of playtesting. The initial version was designed in the style of the MBTI personality test. Feedback from playtesters indicated that this framing felt reductive and lacked professional credibility. We were suggested to use the Big Five test. Rather than presenting the test directly, we developed six teacher identity types that translate Big Five trait combinations into classroom-relevant descriptions through weighted formulas, as shown in the table below. The identity type descriptions, Big Five Formula, and growth edge content were developed by compiling official Big Five documentation, using ChatGPT 5.4 for brainstorming, and critically revising the AI output to achieve language that feels authentic, specific, and useful to a practicing teacher rather than clinical or generic.

Teacher Type	Signature Color	Big Five Formula
Community Energizer	#FF6B5A	$0.40 \times E + 0.40 \times A + 0.20 \times O$
Structured Guide	#2F5D8C	$0.45 \times C + 0.35 \times \text{Emotional Stability} + 0.20 \times A$

Creative Explorer	#8B5CF6	$0.50 \times O + 0.25 \times E + 0.25 \times C$
Steady Supporter	#6BAF8A	$0.45 \times A + 0.40 \times \text{Emotional Stability} + 0.15 \times C$
Reflective Designer	#2A9D8F	$0.45 \times O + 0.40 \times C + 0.15 \times \text{Emotional Stability}$
Quiet Specialist	#5B5F97	$0.40 \times (100 - E) + 0.35 \times C + 0.25 \times O$

Each identity result page presents a short overview with selective bold highlights, natural strengths, and growth edges. Because teachers may retake the assessment at any time, the profile functions as a recurring reflective check-in rather than a one-time classification.

Scenario-based situated practice. The scenario practice system is designed to place teachers inside real, documented classroom situations where facilitation decisions carry distinct pedagogical consequences. Each scenario presents three branching decision points, each with one weak, one partial, and one strong concept-building option; choices map to outcomes from Failure to Success and are followed by a coaching debrief specific to the path taken. The current prototype uses Sean's Number case as a first example: a third-grade mathematics discussion in which a student's claim that six can be both odd and even becomes an opportunity, or a missed opportunity, for collective mathematical reasoning (Mathematics Teaching and Learning to Teach, 2010). The scenario illustrates how the system works, but it is the first of what the platform is designed to become: a library of cases spanning grade levels, subjects, and types of teaching challenge.

The scenario component underwent a major redesign following the second round of playtesting. The initial version used AI-generated classroom scenarios. Playtesters reported that the content felt inconsistent and unstable across attempts, and that the teaching decisions lacked a clear evidential basis, making the coaching debrief feel arbitrary rather than trustworthy. The system was rebuilt using documented classroom transcripts as its foundation. We used AI to help rewrite Sean's Number transcript into a playable branching narrative, then critically revised the output to ensure every decision point, student voice, and mathematical concept remained grounded in the original classroom moment. This makes the debrief specific and credible. It also aligns with a broader insight from Cuban (1990), who observed that professional learning tools frequently fail because they abstract away from the real texture of classrooms, and from Reich and Dukes (2024), who emphasize the value of tools that create conditions for consequential, realistic decision-making rather than idealized simulations.

Formative coaching feedback. Each scenario outcome is followed by a structured debrief with five components: a general outcome summary; focused analysis of how the teacher's choices affected the focal student; analysis of effects on other students in the class; a What you did well card tied to the specific decisions made; and an Improve next time card describing a concrete alternative move. The debrief is written in coaching language that is specific, actionable, and tied to the player's actual decision path rather than generic encouragement. This design is grounded in feedback research: Hattie and Timperley (2007) argue that effective feedback must address where the learner is going, how they are progressing, and what their next steps should be, and that feedback focused on the task and process levels, rather than on praise or personal evaluation, produces the strongest learning gains. Our debrief structure maps directly onto this model: the outcome summary orients the teacher to where their decisions led, the analysis cards explain why, and the Improve next time card provides a concrete next step. The value of this

coaching-centered approach is also supported by evidence from simulation-based teacher training: in a feasibility study of the TeachLive mixed-reality simulator with 62 preservice teachers, Larson et al. (2020) found that coaching feedback was rated the most beneficial aspect of the simulation experience by the largest share of participants, with 86% reporting that the post-simulation coaching was helpful. Teachers are invited to replay the scenario after reading their debrief, enabling iterative experimentation across the outcome branches.

Accessibility design. Our outcome page uses colorblind-friendly, high-contrast colors: green (#2E7D5B) for Success, amber (#B86B00) for Partial Success, and red (#B00020) for Failure. Critically, the design does not rely on color alone. Every outcome state includes a clear uppercase text label (SUCCESS, PARTIAL SUCCESS, FAILURE), a matching icon (checkmark for Success, exclamation mark for Partial Success, warning symbol for Failure), and a distinct visual border treatment, so that the result is fully legible regardless of how a user perceives color. The full game is also keyboard-navigable: all choices, replay buttons, and navigation controls are reachable by Tab and activatable by Enter or Space, using real button elements with a visible focus outline. Screen reader support was added through semantic headings for each major section, aria-label attributes on all choice buttons, and an aria-live region on the result area so that screen readers announce the new outcome when it appears. Finally, every blackboard demonstration includes a short text caption describing the visual content, such as "Six circles are grouped into three pairs of two," so that the mathematical content is accessible without relying solely on the drawing.

4. Timeline of Ideas to Final Product

The development of Teacher Reflection Lab 2.0 moved through three weeks, with two pivotal revisions driven by playtesting: one replacing the MBTI-style personality assessment with a Big Five instrument, and one replacing AI-generated scenario content with a transcript-based system.

Phase	Key Activities & Decisions
Week 1(Ideation +Narrowing)	Initial concept: a web-based teacher development platform with a community case library, AI assistant, and reflective logbook. Scope reduced to a focused MVP: a Big Five personality self-assessment followed by a classroom scenario practice system. Community features and logbook deferred to future iterations.
Week 2(Design +Playtest 1)	Personality assessment designed initially in an MBTI style. First round of playtesting focused on the personality component. Feedback indicated the MBTI framing felt too categorical and lacked research grounding. Assessment redesigned using the Big Five framework with classroom-specific dilemmas. Six teacher identity types developed from weighted Big Five formulas. Scenario system built in its first version using AI-generated classroom scenarios. Feedback identified instability in AI-generated content and a lack of evidential grounding. System redesigned using the documented Sean/Ofala classroom transcript as the prototype case.
Week 3(Build +Playtest 2 +Refinement)	Simulation refined: full-screen blackboard mode for mathematical content, coaching-tone debrief cards, Replay option added. Identity result pages reformatted with short readable paragraphs, green bullet strengths, and amber growth edges. Accessibility improvements implemented: colorblind-friendly

	outcome colors, text labels, and icons added to all performance levels; keyboard navigation and screen reader support integrated throughout.
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The two playtesting-driven pivots were the most consequential design decisions of the project. The first decision to replace MBTI with the Big Five gave the platform greater scientific credibility and made the assessment feel directly relevant to classroom practice. The second decision, replacing AI-generated scenarios with transcript-based cases, resolved both a technical instability problem and a deeper pedagogical one: without a real evidential basis, the coaching debrief could not make specific, trustworthy claims about why each decision mattered. Both revisions reflect a design principle that Gutierrez, Jurow, and Vakil (2020) articulate clearly: the value of an educational tool comes from its fidelity to real contexts, not from its technological novelty.

5. Proposed Improvements

Several meaningful improvements would be made in a full revision of Teacher Reflection Lab 2.0, spanning the scenario library, the recommendation engine, the identity type system, and future platform features.

The highest priority improvement is expanding the scenario library and implementing the recommendation engine. The current prototype demonstrates the scenario system with one case; the full vision is a library of documented classroom cases spanning different grade levels, subject areas, and types of teaching challenges. The recommendation engine would use each teacher's identity type scores and self-identified growth areas to surface the most relevant scenarios. For example, a Community Energizer, whose growth edge involves protecting processing time for quieter students, might be directed toward scenarios involving extended wait-time decisions; a Quiet Specialist, whose growth edge involves making thinking visible, might be directed toward cases requiring public whole-class facilitation.

The six identity types would also benefit from iterative co-design with practicing teachers. The current type descriptions were developed by drawing on the Big Five literature and classroom intuition. A stronger version would involve testing with K-12 teachers from diverse school contexts to confirm that the identity overviews feel authentic, that the strengths resonate without feeling hollow, and that the growth edges feel actionable. Gutierrez, Jurow, and Vakil (2020) argue that educational designs should remain responsive to local context and community knowledge; co-design is the appropriate path toward meeting that standard.

Looking further ahead, we envision expanding Teacher Reflection Lab 2.0 into a more comprehensive self-reflection platform. Planned future features include a personal logbook where teachers can record and revisit reflections after each scenario attempt, tracking how their thinking evolves over time; persistent personality profiles that teachers can retake and compare across career stages; and expanded privacy controls ensuring all data remains entirely in the teacher's own hands. These features would transform the current two-component prototype into a longitudinal professional development journal built around the teacher's own growth.

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